

232.2.4971 - Quantum computation

Homework Assignment 01

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Due: 09/05/2026 - extended

Question 01 - Inner Products, Angles and Measurement of a single qubit

Let $|\psi\rangle = \cos(\pi/8)|0\rangle + \sin(\pi/8)|1\rangle$.

- Write the state $|\psi\rangle$ using the standard notation. Write $\langle\psi|$ using the standard notation.
- Verify that $\| |\psi\rangle \|^2 \equiv \langle\psi|\psi\rangle = 1$.
- What is the inner product between $|\psi\rangle$ and $|0\rangle$? What is the angle between $|\psi\rangle$ and $|0\rangle$? (Note: the angle between two vectors u, v is defined as $\arccos\left(\frac{\langle u, v \rangle}{\|u\| \cdot \|v\|}\right)$)
- You are given a quantum state which is either $|\psi\rangle$ or $|0\rangle$ uniformly at random. You win if you are able to identify correctly the state that you are given, and you lose otherwise. Calculate the winning probabilities of the following strategies:
 - Measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.
 - Apply the Hadamard gate $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, and then measure the qubit in the computational basis. Answer '0' if the outcome is '0', and answer '1' if the outcome is '1'.

Answer

- in standard notation $|\psi\rangle$ is $\begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix}$, and $\langle\psi|$ is $\begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix}$.

- Let's verify $\| |\psi\rangle \|^2 = 1$:

$$\| |\psi\rangle \|^2 = \langle\psi|\psi\rangle = \begin{bmatrix} \cos(\pi/8) & \sin(\pi/8) \end{bmatrix} \begin{bmatrix} \cos(\pi/8) \\ \sin(\pi/8) \end{bmatrix} = \cos^2(\pi/8) + \sin^2(\pi/8) = 1$$

- The inner product between $|\psi\rangle$ and $|0\rangle$ is:

$$\begin{aligned} \langle\psi|0\rangle &= \cos(\pi/8) \langle 0|0\rangle + \sin(\pi/8) \langle 0|1\rangle \\ &= \cos(\pi/8) \end{aligned}$$

The angle is:

$$\begin{aligned} \frac{\langle\psi|0\rangle}{\| |\psi\rangle \| \| |0\rangle \|} &= \frac{\cos(\pi/8)}{|\langle\psi|\psi\rangle|^2 \cdot |\langle 0|0\rangle|^2} \\ &= \frac{\cos(\pi/8)}{|1|^2 \cdot |1|^2} \\ &= \cos(\pi/8) \end{aligned}$$

- (a) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned} \Pr\{win\} &= \Pr\{win | |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win | |\psi\rangle\} \Pr\{|\psi\rangle\} \\ &= \langle 0|\Pi_0|0\rangle / 2 + \langle\psi|\Pi_1|\psi\rangle / 2 \\ &= |\langle 0|0\rangle|^2 / 2 + |\langle 1|\psi\rangle|^2 / 2 \\ &= (1 + \sin^2(\pi/8)) / 2 \approx 0.57 \end{aligned}$$

- (b) Let Π_0 and Π_1 be the projection operator for the computational basis, then the probability of winning is:

$$\begin{aligned}
 \Pr\{win\} &= \Pr\{win \mid |0\rangle\} \Pr\{|0\rangle\} + \Pr\{win \mid |\psi\rangle\} \Pr\{|\psi\rangle\} \\
 &= \langle 0 | H \Pi_1 H | 0 \rangle / 2 + \langle \psi | H \Pi_0 H | \psi \rangle / 2 \\
 &= (\langle 0 | + \langle 1 |) | 1 \rangle \langle 1 | (| 0 \rangle + | 1 \rangle) / 4 \\
 &\quad + ((\cos(\pi/8) + \sin(\pi/8)) \langle 0 | + (\cos(\pi/8) - \sin(\pi/8)) \langle 1 |) \\
 &\quad | 0 \rangle \langle 0 | \\
 &\quad (\cos(\pi/8) + \sin(\pi/8)) | 0 \rangle + (\cos(\pi/8) - \sin(\pi/8)) | 1 \rangle) / 4 \\
 &= |\langle 1 | 1 \rangle|^2 / 4 + |(\cos(\pi/8) + \sin(\pi/8)) \langle 0 | 0 \rangle|^2 / 4 \\
 &= (1 + 1 + 2 \cos(\pi/8) \sin(\pi/8)) / 4 \\
 &= 1/2 + \sin(\pi/4) / 4 \approx 0.68
 \end{aligned}$$

We can see that this strategy is better than the previous one.

Question 2 - Tensor Product State

Let $|\alpha\rangle = a|0\rangle + b|1\rangle$, $|\beta\rangle = c|0\rangle + d|1\rangle$, where $a, b, c, d \in \mathbb{C}$ (Notice that this vector may not be normalized).

- Write the state $|\alpha\beta\rangle$ in the $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ basis, using both the Dirac notation and the usual matrix notation.
- Calculate the standard inner product between $|\alpha\rangle$ and $|\beta\rangle$.
- Calculate the norm of $|\alpha\rangle$ and prove that $\| |\alpha\beta\rangle \|^2 = \| |\alpha\rangle \|^2 \cdot \| |\beta\rangle \|^2$.
- Prove or disprove: Is the state $|\psi\rangle = \sqrt{\frac{1}{3}}|01\rangle + \sqrt{\frac{2}{3}}|10\rangle$ a tensor product state?
- Same question as previous item, but for the state $|\psi\rangle = \sqrt{\frac{1}{3}}|11\rangle + \sqrt{\frac{2}{3}}|10\rangle$.

Answer

- a) In Dirac notation:

$$\begin{aligned}
 |\alpha\beta\rangle &= (a|0\rangle + b|1\rangle)(c|0\rangle + d|1\rangle) \\
 &= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle
 \end{aligned}$$

In matrix notation:

$$\begin{aligned}
 |\alpha\beta\rangle &= \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} \\
 &= \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}
 \end{aligned}$$

- b) The inner product is:

$$\begin{aligned}
 \langle \alpha | \beta \rangle &= (a^* \quad b^*) \begin{pmatrix} c \\ d \end{pmatrix} \\
 &= a^*c + b^*d
 \end{aligned}$$

- c) The norm of $|\alpha\rangle$ is:

$$\begin{aligned}
 \| |\alpha\rangle \| &= \sqrt{\langle \alpha | \alpha \rangle} \\
 &= \sqrt{(a^* \quad b^*) \begin{pmatrix} a \\ b \end{pmatrix}} \\
 &= \sqrt{|a|^2 + |b|^2}
 \end{aligned}$$

prove that $\|\alpha\beta\rangle\|^2 = \|\alpha\rangle\|^2 \cdot \|\beta\rangle\|^2$:

$$\begin{aligned}\|\alpha\beta\rangle\|^2 &= \langle\alpha\beta|\alpha\beta\rangle \\ &= \langle\alpha|\alpha\rangle \langle\beta|\beta\rangle \\ &= \|\alpha\rangle\|^2 \cdot \|\beta\rangle\|^2\end{aligned}$$

d) This is not a tensor product state:

A tensor product state is of the form $|\alpha\beta\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$, therefore for $|\psi\rangle$ to be a tensor product state the following must hold:

$$\begin{aligned}ac &= 0 \\ ad &= \sqrt{\frac{1}{3}} \\ bc &= \sqrt{\frac{2}{3}} \\ bd &= 0\end{aligned}$$

There is no solution for this set of equations since from $ac = 0$ either a or c is 0 but this contradicts the fact that both ad and bc are not 0 item[e)] This is a tensor product state:

A tensor product state is of the form $|\alpha\beta\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$, therefore for $|\psi\rangle$ to be a tensor product state the following must hold:

$$\begin{aligned}ac &= 0 \\ ad &= 0 \\ bc &= \sqrt{\frac{2}{3}} \\ bd &= \sqrt{\frac{1}{3}}\end{aligned}$$

a possible solution for this set of equations is: $\begin{matrix} a=0 \\ b=1 \\ c=\sqrt{\frac{2}{3}} \\ d=\sqrt{\frac{1}{3}} \end{matrix}$ and therefore this state is a tensor product state.

1 Question 3 - Operations and Measurements of Multiple Qubits